



AI JOURNAL WITH SENTIMENT ANALYSIS

SPRINT 3

15 May, 2026

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TEAM NEXUS



KRISHNA
Scrum Master |
Developer



CHINTAN KOTADIYA
Developer | Tester



RAFAELA HOLLANDA
Team Leader |
Developer



VANESSA AGBUGBA
Front End Developer



SANKET MATROJA
AI Generalist



SAI
Developer

TEAM MEMBERS

IMPROVEMENTS MADE FROM PROFESSOR'S FEEDBACK

[Assignments](#) > View Feedback

Feedback for Sprint 2 (Deliverable 2)

Submission Feedback

Overall Feedback

No feedback for you guys because you guys are awesome.

Score

99 / 100 - 99 %

Feedback Date

Apr 22, 2026 6:56 PM

PROJECT DESCRIPTION

Project Name:	Mind Mirror
Team:	Nexus
Project Description:	For individuals who struggle to understand and manage their emotions, the AI-powered journaling application is a smart, emotion-tracking tool that provides real-time sentiment analysis, mood visualization, and actionable insights. Unlike traditional journaling apps that lack emotional analytics or structure, our application combines journaling with AI-driven insights to help users recognize patterns, prevent burnout, and improve emotional well-being.
Benefit Outcomes:	Users can anticipate measurable improvements in emotional awareness, reflection consistency, and stress management. The application is expected to increase emotional recognition by 60%, reduce stress levels by 40%, and triple journaling consistency through its quick-entry and real-time feedback features.
Github Link:	https://github.com/htmw/2026S-Nexus

TEAM WORKING AGREEMENT

COMMUNICATION PLAN

- Primary instant meeting platform: WhatsApp
- Video Meeting: Discord

Team members are expected to:

- Communicate respectfully and professionally
- Respond to messages within a reasonable time frame (ideally within 24 hours)
- Share progress updates, blockers, and concerns openly
- Use the agreed communication tools for all project-related discussions

MEETINGS & AVAILABILITY

The team will hold three regular sync meetings per week:

- Sunday: 10:00 AM • Tuesday: 6:30 PM • Thursday: 9:00 PM

Additional meetings may be scheduled as needed based on sprint requirements. Expectations:

- All members should attend scheduled meetings
- If unable to attend, members must notify the team in advance
- Meeting notes and action items will be documented and shared

TEAM WORKING AGREEMENT

TASK MANAGEMENT

- Task tracking tool: Jira

All project tasks will be:

- Clearly defined and assigned
- Tracked through Jira
- Updated regularly to reflect progress
- Completed before agreed internal deadlines

CONFLICT RESOLUTION

In the event of conflict:

1. The concerned party should first express the issue to the team openly and respectfully
2. The team will attempt to resolve the issue collaboratively
3. If unresolved, the issue will be escalated to the Team Leader / Scrum Coordinator
4. Further escalation (e.g., to the instructor) will be considered only if necessary

TEAM WORKING AGREEMENT

WORK EXPECTATIONS & DEFINITION OF DONE

Each task is considered “Done” when:

- It meets the agreed acceptance criteria
- It is reviewed by at least one other team member (when applicable)
- Code runs without critical errors
- Relevant documentation is updated
- Changes are committed to the shared repository

Team members are responsible for:

- Completing assigned work on time
- Communicating blockers early
- Supporting teammates when needed

STUDENT PERSONA



NAME: SARAH MARTINEZ

Age: 21

Gender: Female

Occupation: Undergraduate Computer Science Student

Sarah is a full-time college student balancing coursework, part-time work, and social commitments. During exam periods, her stress levels rise significantly, but she often ignores the early warning signs of burnout. She occasionally journals but finds long entries overwhelming and inconsistent. She wants a simple way to reflect daily without spending too much time.

Challenges:

- Experiences fluctuating stress levels during the semester
- Inconsistent journaling habits
- Difficulty recognizing emotional patterns over time
- Feels overwhelmed during midterms and finals

Goals:

- Track emotional trends throughout the semester
- Identify early signs of burnout
- Receive short, actionable suggestions
- Maintain better emotional balance during academic pressure

WORKING PROFESSIONAL PERSONA



NAME: DAVID CHEN

Age: 32

Gender: Male

Occupation: Software Engineer

David works remotely for a tech company and spends most of his day attending meetings and coding. While he performs well professionally, he often feels mentally drained without clearly understanding the cause. He values productivity and self-improvement but rarely pauses to reflect on his emotional state.

Challenges:

- Blurred boundaries between work and personal life
- Emotional fatigue during high-pressure deadlines
- No structured way to monitor mood patterns
- Limited time for long reflection exercises

Goals:

- Understand how work stress affects his mood
- Monitor emotional stability over time
- Receive quick, data-driven insights
- Improve work-life balance

WORKING PARENT PERSONA



NAME: EMILY PETERSON

Age: 40

Gender: Female

Occupation: HR Manager

Emily balances a demanding job with parenting responsibilities. Her schedule leaves little time for self-reflection, yet she recognizes the importance of emotional awareness. She often feels emotionally exhausted but cannot clearly identify triggers or patterns. She prefers simple, easy-to-use digital tools.

Challenges:

- Limited time for lengthy journaling
- Emotional fatigue from multitasking responsibilities
- Difficulty identifying emotional triggers
- Wants non-clinical, supportive guidance

Goals:

- Build daily emotional awareness
- Identify patterns in stress levels
- Receive manageable well-being suggestions
- Maintain emotional stability for herself and her family

SELF-IMPROVEMENT ENTHUSIAST PERSONA



NAME: ALEX RIVERA

Age: 27

Gender: Male

Occupation: Fitness Coach

Alex actively tracks his workouts, sleep cycles, and productivity habits. He believes emotional well-being is just as important as physical health but lacks a structured system to measure it. He is interested in data-driven insights and enjoys tools that provide analytics and visualizations.

Challenges:

- Journaling apps provide no analytical insights
- No measurable way to track emotional consistency
- Wants deeper pattern recognition
- Prefers technology-backed recommendations

Goals:

- Quantify emotional trends
- Compare mood ratings with AI sentiment analysis
- Identify emotional volatility
- Optimize overall mental performance

TECHNOLOGIES

FRONT END



BACK END



Hugging Face

DATABASE



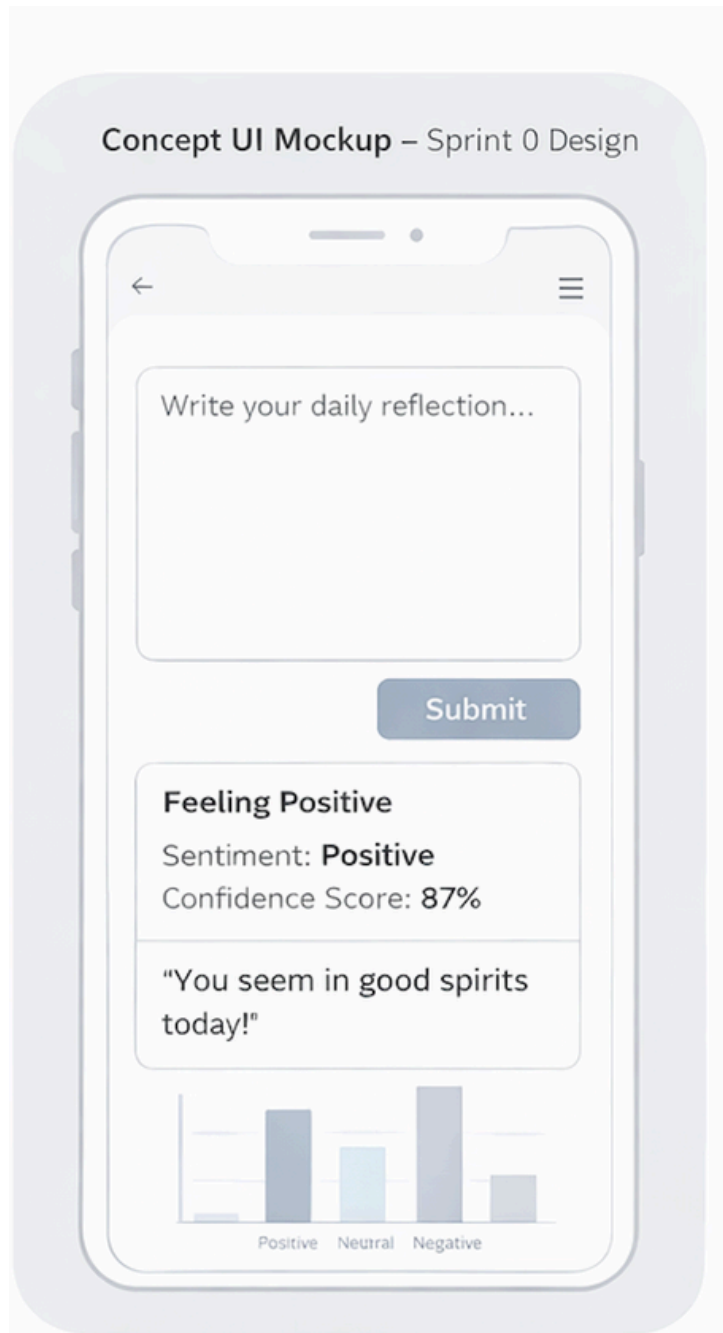
TOOLS



Canva



MINIMAL VIABLE PRODUCT (MVP)



An AI-powered journaling application with real-time emotional analysis and visual mood tracking.

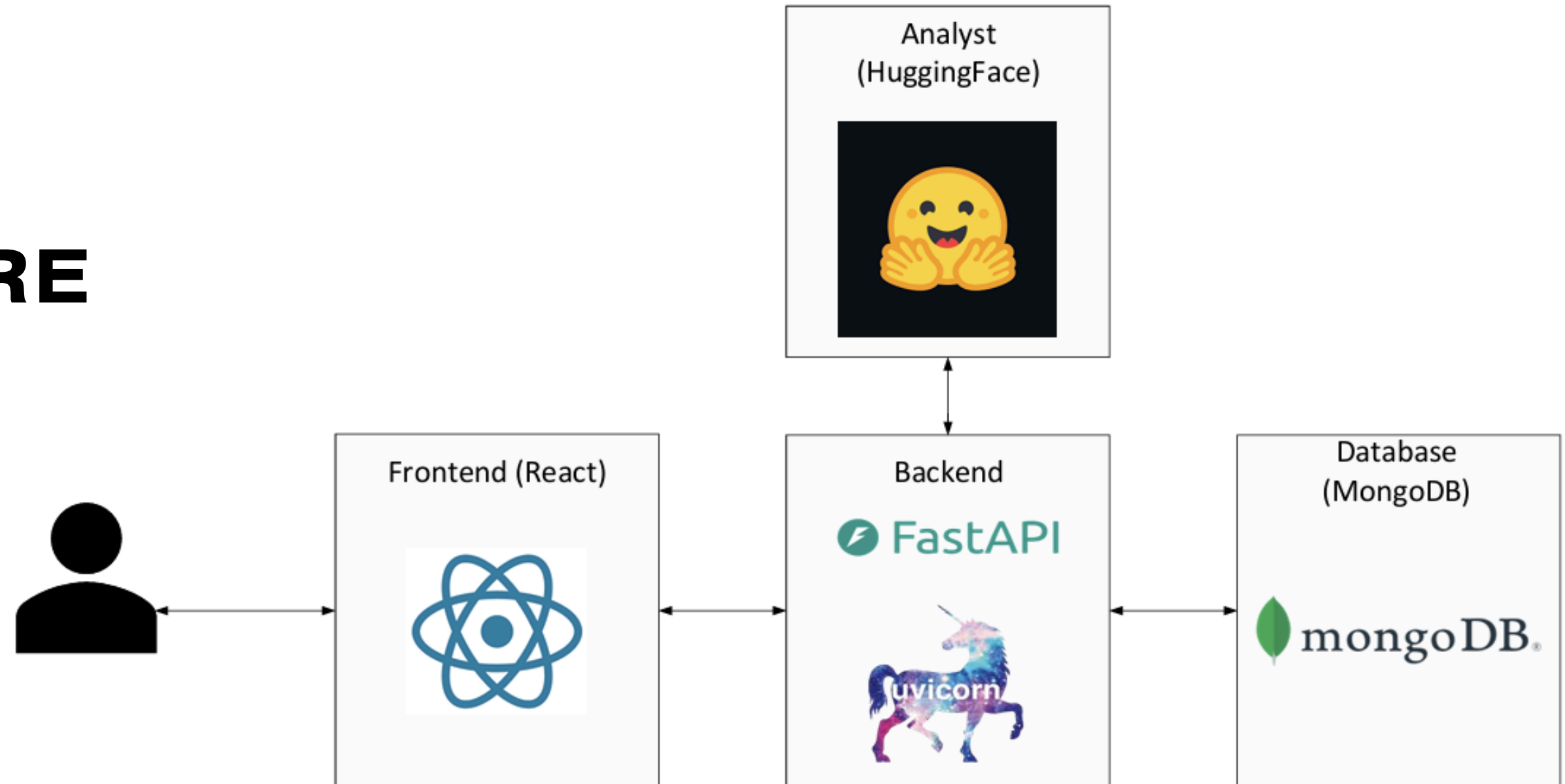
- **Real-Time Sentiment Analysis** – Journal entries are analyzed instantly using an NLP model.
- **Mood Classification System** – Each entry is labeled as Positive, Neutral, or Negative.
- **Confidence Score Display** – Shows the model's certainty level for each classification.
- **Immediate Mood Feedback** – Provides short, supportive emotional insights after analysis
- **Visual Mood Distribution** – Displays a basic chart showing mood trends over time.
- **Local Entry Persistence** – Saves journal entries locally for later review (MVP scope).

COMPONENTS

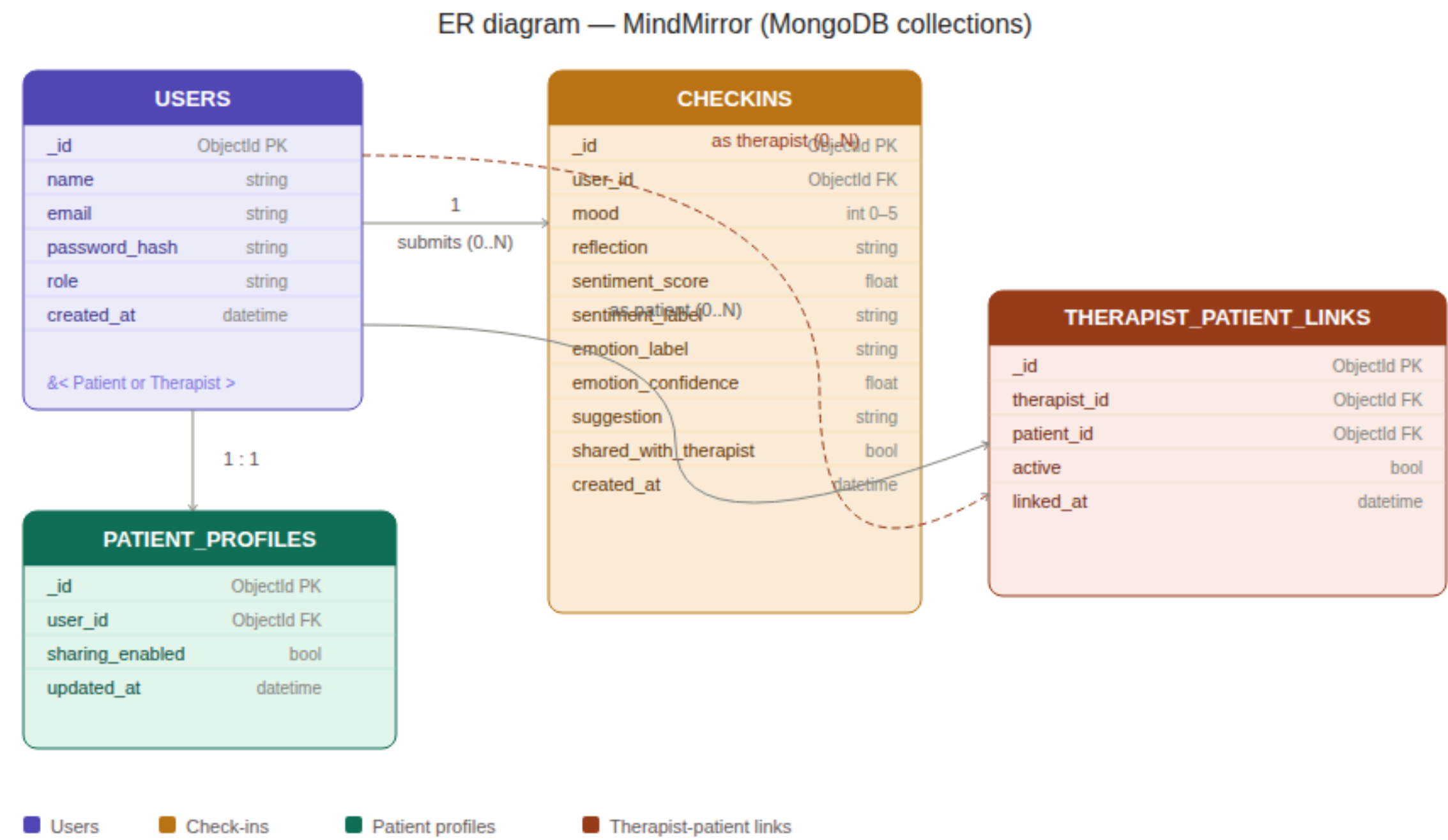
- **Journal Input Manager** : Handles user journal entry input and submission through the web interface.
- **Backend API Manager** : Manages HTTP requests and communication between frontend, sentiment model, and database.
- **Sentiment Analysis Manager** : Processes journal text and classifies sentiment (Positive / Neutral / Negative).
- **Confidence Scoring Module**: Extracts prediction probability scores from the DistilBERT output.
- **Mood Feedback Generator**: Converts sentiment results into short, supportive emotional insights.
- **Data Persistence Layer** : Stores journal entries, sentiment labels, confidence scores, and timestamps.
- **Mood Visualization Manager**: Retrieves stored data and generates mood distribution charts over time.



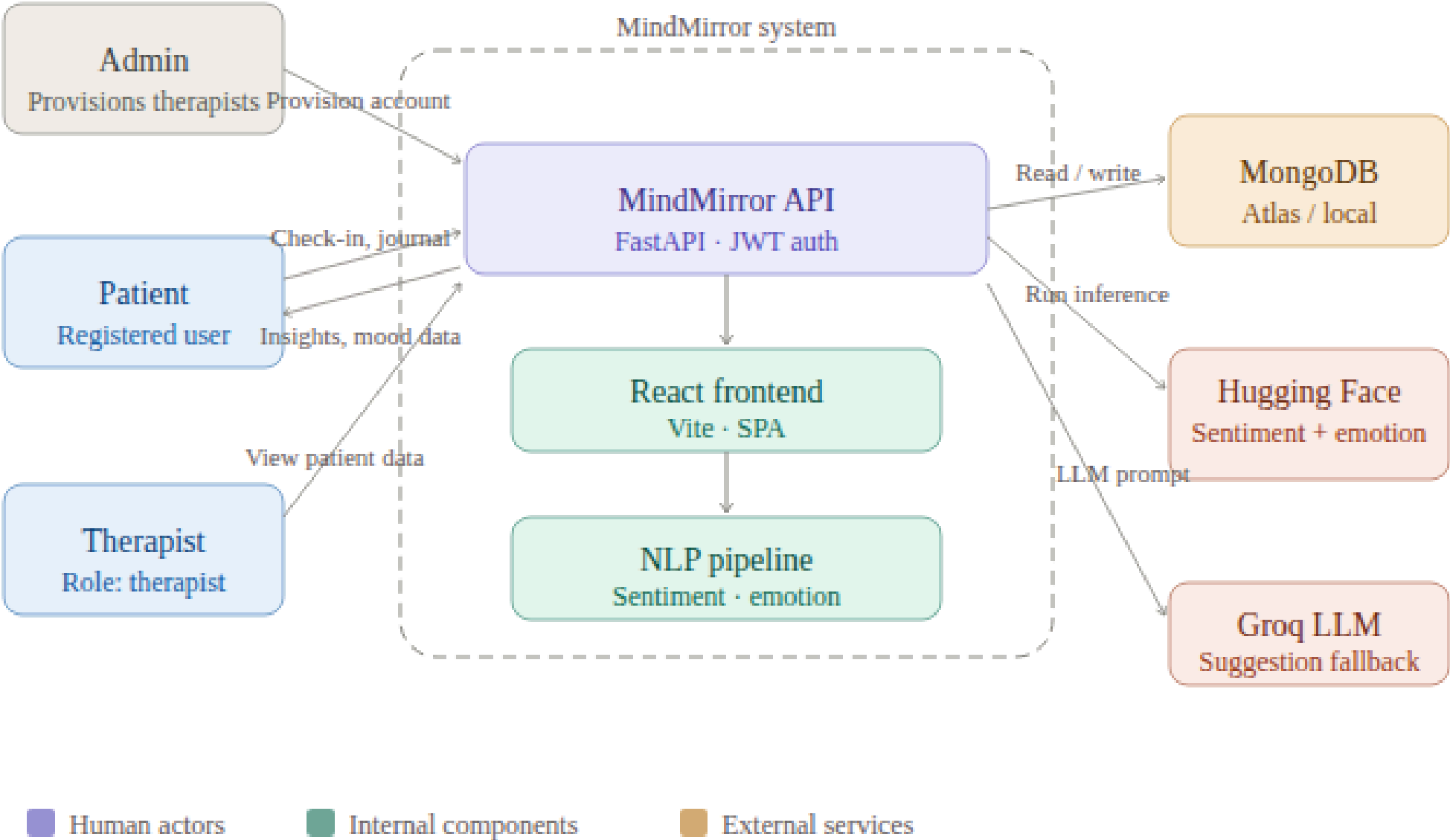
ARCHITECTURE DIAGRAM



ER DIAGRAM

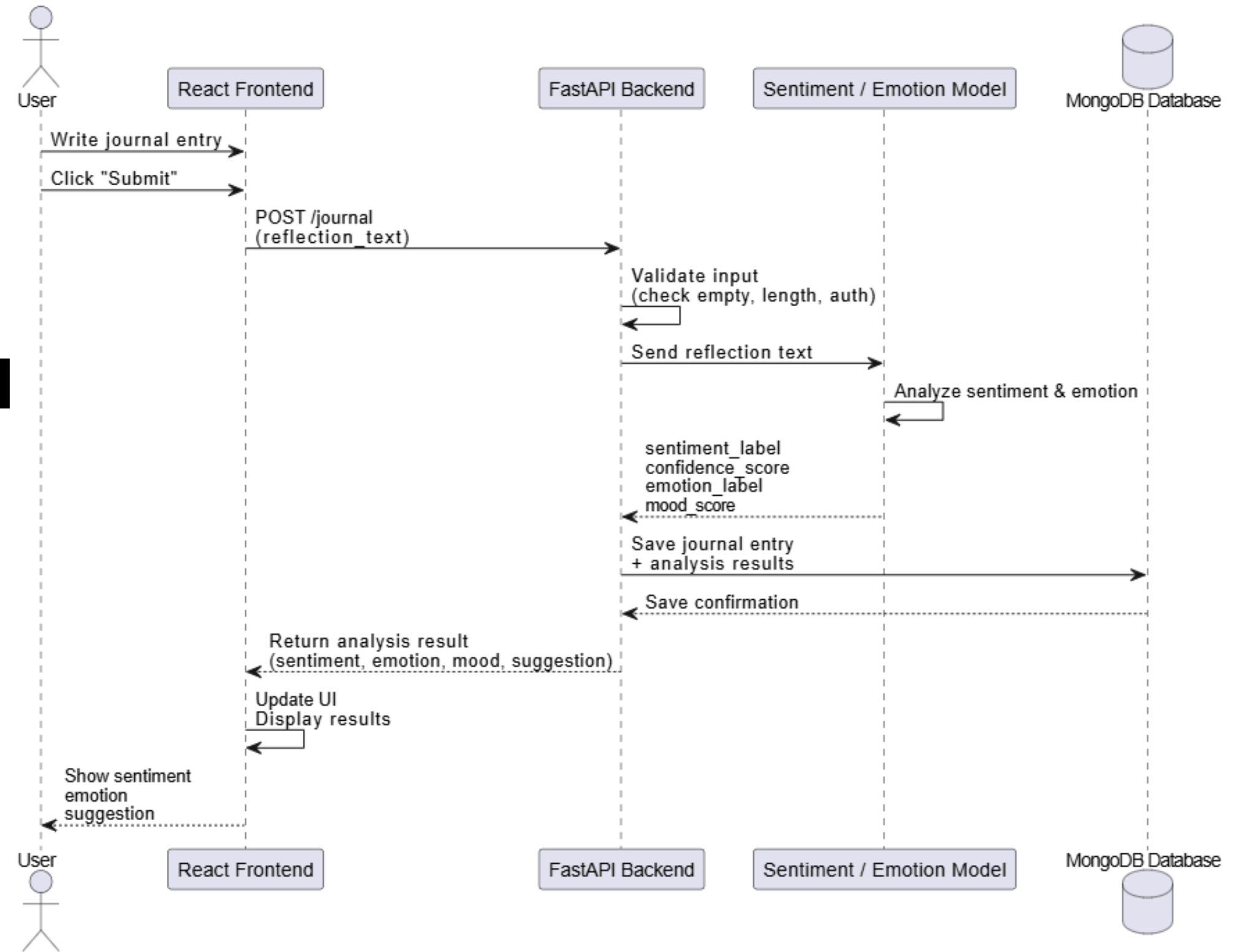


CONTEXT DIAGRAM



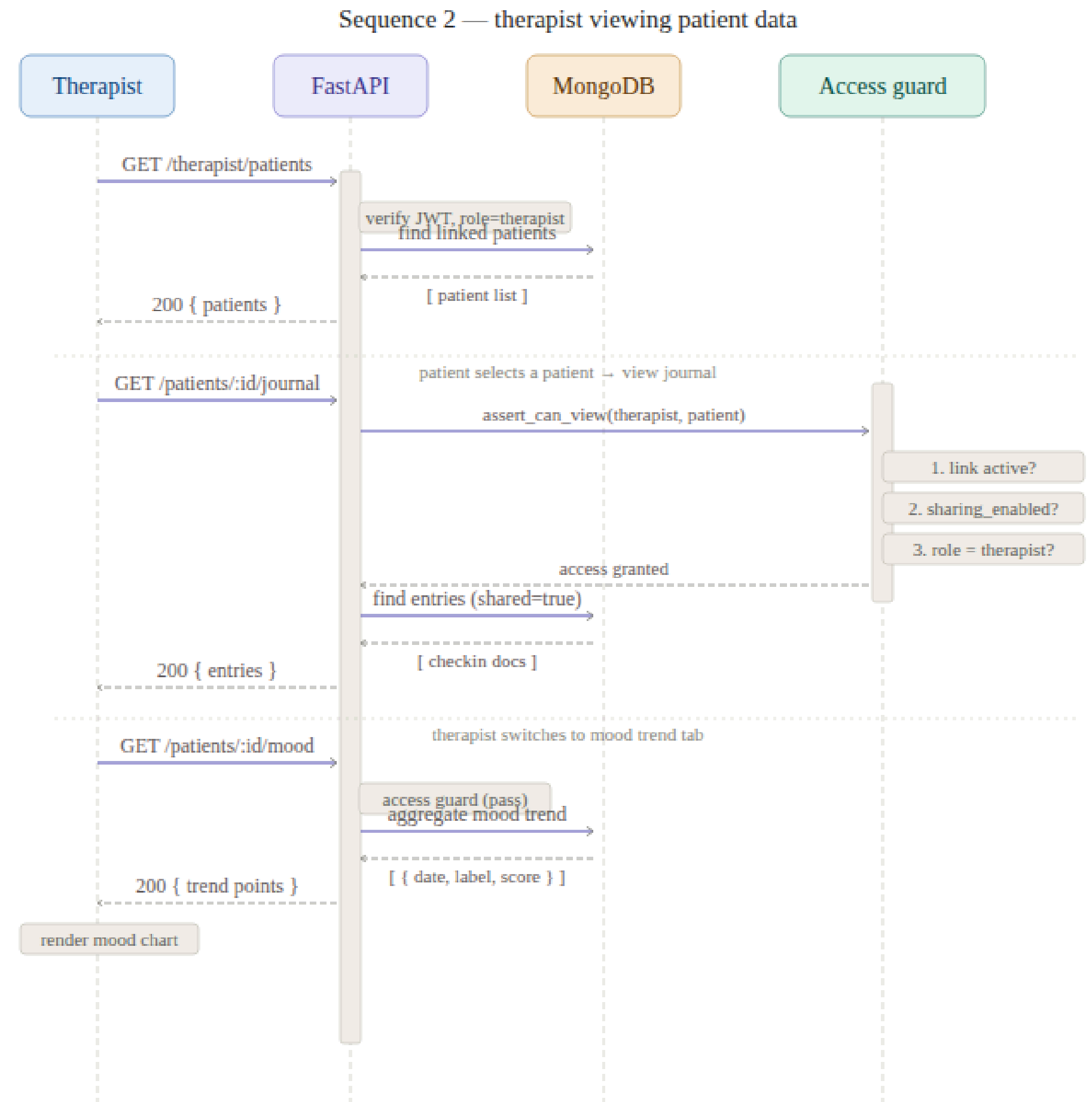
SEQUENCE DIAGRAM

-JOURNAL ENTRY



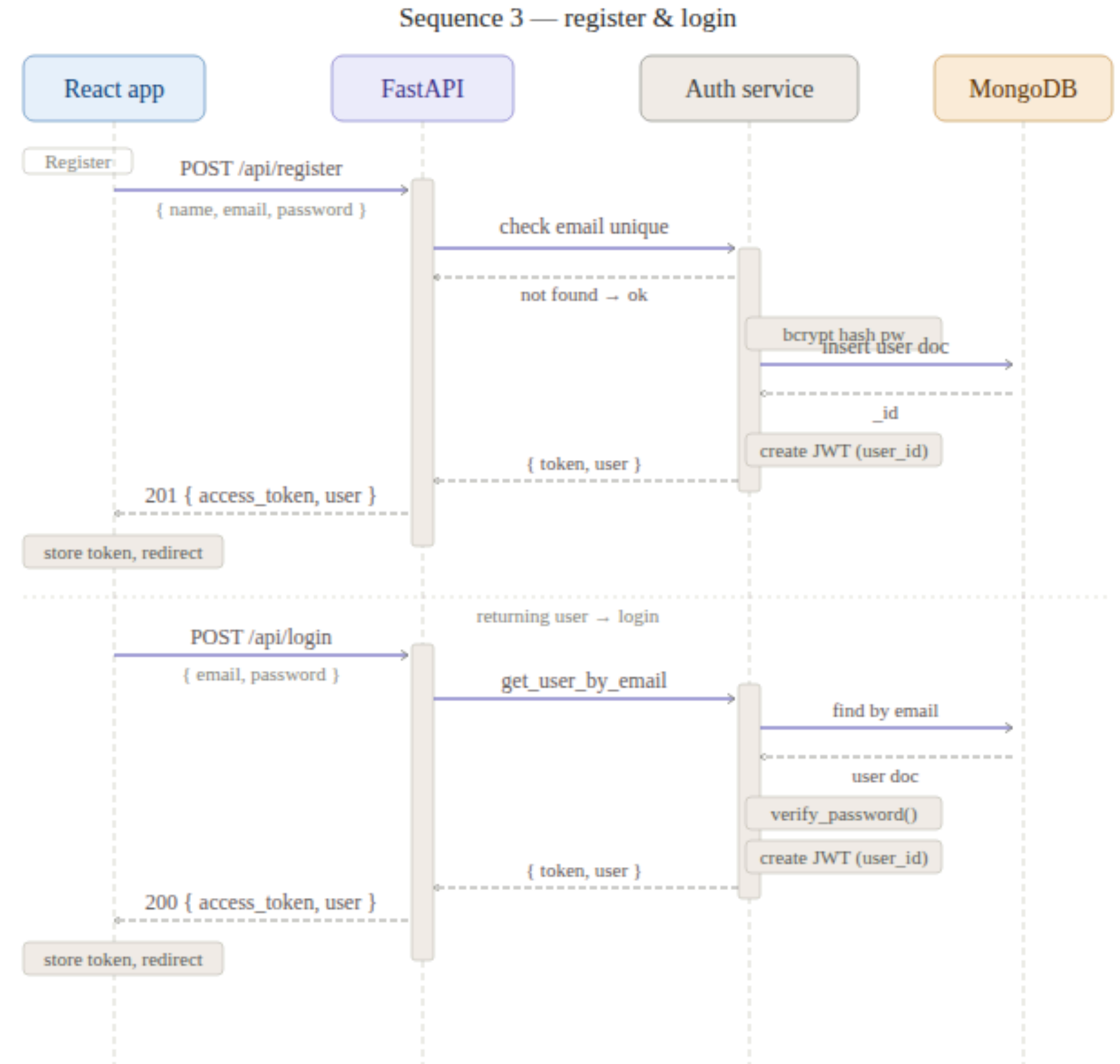
SEQUENCE DIAGRAM

-THERAPIST



SEQUENCE DIAGRAM

-AUTH



SPRINT 2 RECAP

What We Set Out To Do

- Deliver a usable authenticated patient flow with sign up, sign in, protected routes, and logout.
- Extend the app beyond submission into Past Entries, Insights, and mood-over-time support.
- Store sentiment results on entries so mood history and charting use persisted data.
- Improve UX reliability with better loading states, safer error handling, and protected APIs.

What We Delivered

- Implemented signup, login, JWT auth, auth headers, protected routes, and logout flow.
- Added Past Entries UI, Insights API support, mood history support, and updated mood chart behavior.
- Integrated therapist-facing data access and patient-view groundwork on the integration branch.
- Added global backend error handling, frontend API interception, toast alerts, and graceful ML failure fallback.

PRODUCT BACKLOG

THERAPIST ACCESS & SHARING

US20 – Therapist Journal Access

As a user, I want my therapist to have access to my journal and mood data so that they can better understand my emotional state.

Acceptance Criteria

- System supports linking a user to a therapist account
- Therapist role exists in the system
- Therapist can access user data

Priority: Must

Story Points: 5

US21 – Enable Journal Sharing

As a user, I want to enable sharing of my journal and mood data so that my therapist can access it.

Acceptance Criteria

- A "Share toggle is visible in user settings
- When toggled ON: sharing state saved in backend
- Therapist gains access to journal + mood data

Priority: Must

Story Points: 8

US22 – Disable Sharing

As a user, I want to disable sharing at any time so that I can revoke my therapist's access.

Acceptance Criteria

- User can toggle sharing OFF at any time
- Backend updates immediately
- Therapist loses access instantly
- Previously visible data is no longer accessible

Priority: Could

Story Points: 5

THERAPIST DASHBOARD

US23 – View Assigned Users

As a therapist, I want to see my assigned users so that I can access their shared data.

Acceptance Criteria

- Therapist dashboard displays a list of assigned users
- Each user shows basic identifier
- Only users linked to that therapist are shown

Priority: Could

Story Points: 5

US24 – View User Journal & Mood

As a therapist, I want to view journal entries and mood trends of users who have enabled sharing so that I can understand their emotional patterns.

Acceptance Criteria

- Therapist can view journal entries in chronological order
- Mood trends shown as chart or list
- Data is shown only if sharing is ON

Priority: Could

Story Points: 5

[Detailed spreadsheet here](#)

SPRINT 1 SUMMARY

Sprint Name: Sprint 1

Sprint Duration: February 20th - March 26th *

Sprint Length: 1 month *

Sprint Goal: This sprint focused on delivering a functional journaling workflow that allows users to submit journal entries and begin sentiment analysis processing.

Stories Committed: US1, US2, US3, US4, US5, US6, US7, US8, US9, US10, US13

Total Story Points Committed: 41

SPRINT 2 SUMMARY

Sprint Name: Sprint 2

Sprint Duration: April 3rd - April 16th

Sprint Length: 2 weeks

Sprint Goal: This sprint focused on incorporating users into the system by implementing secure registration, login, and authenticated access to journal features.

Stories Committed: US6, US11, US12, US13, US15, US16, US17, US18, US19.

Total Story Points Committed: 40

SPRINT 3 BACKLOG

Sprint Name: Sprint 3

Sprint Duration: April 17th - May 15th

Sprint Length: 3 weeks

Sprint Goal: This sprint focuses on enabling secure, user-controlled sharing of journal and mood data between users and their therapists, including the ability to grant and revoke access at any time.

Stories Committed: US20, US21, US22, US23, US24

Total Story Points Committed: 28

SPRINT 3 TASK BREAKDOWN

Sprint 3 focused on therapist sharing workflows. User stories were broken into smaller technical tasks so ownership, implementation effort, and dependencies could be tracked clearly across backend, frontend, and integration work.

Why Task Breakdown Mattered

- Clarified who owned each part of therapist linking, sharing, and read-only data access.
- Separated backend API work from frontend workflow and UI tasks.
- Made it easier to track which therapist-sharing stories were complete versus still in progress.
- Supported parallel work on access control, dashboard views, sharing toggles, and patient detail flows.

SPRINT 3 TEST CASE

Sprint 3							
US20 — As a user, I want my therapist to have access to my journal and mood data so that they can better understand my emotional state.							
Acceptance criteria: System supports linking a user to a therapist account; therapist role exists in the system; therapist can potentially access user data controlled by sharing setting; feature is visible in UI.							
TC ID	Scenario	Type	Status	Description	Input / Payload	Expected Result	Notes
TC_US20_01	Valid flow	Integration	PASS	User links to a valid therapist account	Valid patient account + valid therapist identifier	Link created successfully; therapist association stored	Happy path linking
TC_US20_02	Invalid input	Integration	PASS	User tries to link to an invalid or non-therapist account	Invalid therapist ID / patient ID	Request rejected with clear error	Therapist role validation
TC_US20_03	System response	Integration	PASS	Therapist link is reflected in UI or settings	Existing linked therapist	Linked therapist details visible in UI	Visibility check
TC_US20_04	Edge case	Integration	PASS	User attempts duplicate linking to the same therapist	Same therapist linked twice	Duplicate link prevented	Unique link protection
Totals			4	0	0		

[Detailed spreadsheet here](#)

SPRINT 3 TEST CASE

US22 — As a user, I want to disable sharing at any time so that I can revoke my therapist’s access.

Acceptance criteria: User can toggle sharing OFF at any time; backend updates immediately; therapist loses access instantly; previously visible data is no longer accessible; UI confirms the change.

TC ID	Scenario	Type	Status	Description	Input / Payload	Expected Result	Notes
TC_US22_01	Valid flow	Integration	PASS	User disables sharing from enabled state	Toggle OFF	Sharing set to OFF in backend	Happy path revoke access
TC_US22_02	System response	Integration	PASS	Therapist immediately loses access after sharing is disabled	Therapist requests patient data after revoke	HTTP 403 / access denied	Immediate revocation
TC_US22_03	System response	Integration	PASS	UI reflects disabled sharing state	Sharing turned OFF	Toggle updates correctly in UI	UI confirmation
TC_US22_04	Edge case	Integration	PASS	Previously visible data is no longer accessible after revocation	Therapist revisits patient detail	No journal or mood data shown	Read access removed
Totals			4	0	0		

[Detailed spreadsheet here](#)

SPRINT 3 TEST CASE

Sprint 3

US24 — As a therapist, I want to view journal entries and mood trends of users who have enabled sharing so that I can understand their emotional patterns.

Acceptance criteria: Therapist can select a user from their list; therapist can view journal entries in chronological order and mood trends; data is shown only if sharing is ON; if sharing is OFF show a friendly message; therapist cannot edit or delete any data.

TC ID	Scenario	Type	Status	Description	Input / Payload	Expected Result	Notes
TC_US24_01	Valid flow	Integration	PASS	Linked therapist views patient journal entries with sharing enabled	GET /api/therapist/patients/:patientId/journal with therapist auth + linked patient	HTTP 200; journal entries returned in chronological order	Happy path journal access
TC_US24_02	Valid flow	Integration	PASS	Linked therapist views patient mood trend data with sharing enabled	GET /api/therapist/patients/:patientId/mood with therapist auth + linked patient	HTTP 200; mood trend points returned for charting	Happy path mood access
TC_US24_03	Invalid input	Integration	PASS	Therapist requests patient data when sharing is disabled	Therapist auth + linked patient with sharing OFF	HTTP 403; friendly not-shared response returned	Sharing-off enforcement
TC_US24_04	Invalid input	Integration	PASS	Unlinked therapist attempts to view patient data	Therapist auth + unrelated patient ID	HTTP 403; access denied	Link validation
TC_US24_05	System response	Integration	PASS	Therapist attempts edit or delete action on patient data	Mutating request from therapist role on patient data endpoint	HTTP 403; write action rejected	Read-only protection
Totals			5	0	0		

[Detailed spreadsheet here](#)

SPRINT 3 TEST CASE SUMMARY

Sprint 3 centered on therapist-sharing workflows, revoking access, and therapist-side visibility into shared patient data.

Sprint Focus

- Expand the product beyond individual patient use into therapist-linked access and controlled data sharing.
- Support secure enable/disable sharing flows so patients control when therapists can access their data.
- Add therapist-facing views for assigned users, journal history, and mood trends while keeping the experience read-only.
- Prepare process deliverables alongside implementation work, including retrospective, planning, technical paper, and user manual.

SPRINT 3 STORIES COMPLETED



US20 Therapist access setup and therapist-linked workflow support



US22 Disable sharing and revoke therapist access immediately

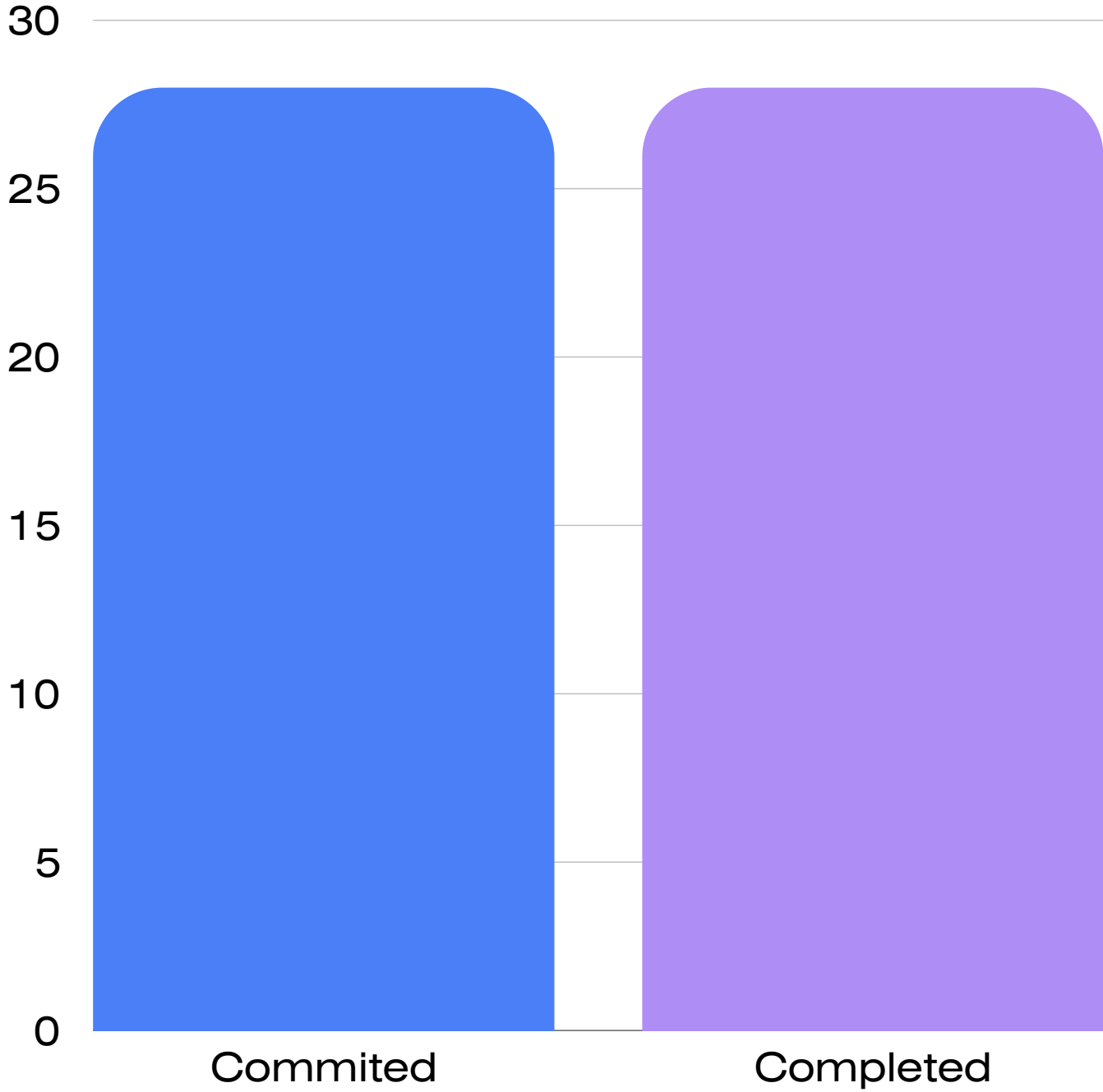


Therapist journal and mood-trend viewing with sharing and read-only protection

METRICS

Team Velocity

Team estimated points	28
Team points delivered	28
Total points carried over	28
Team Velocity	100%



METRICS AVERAGE

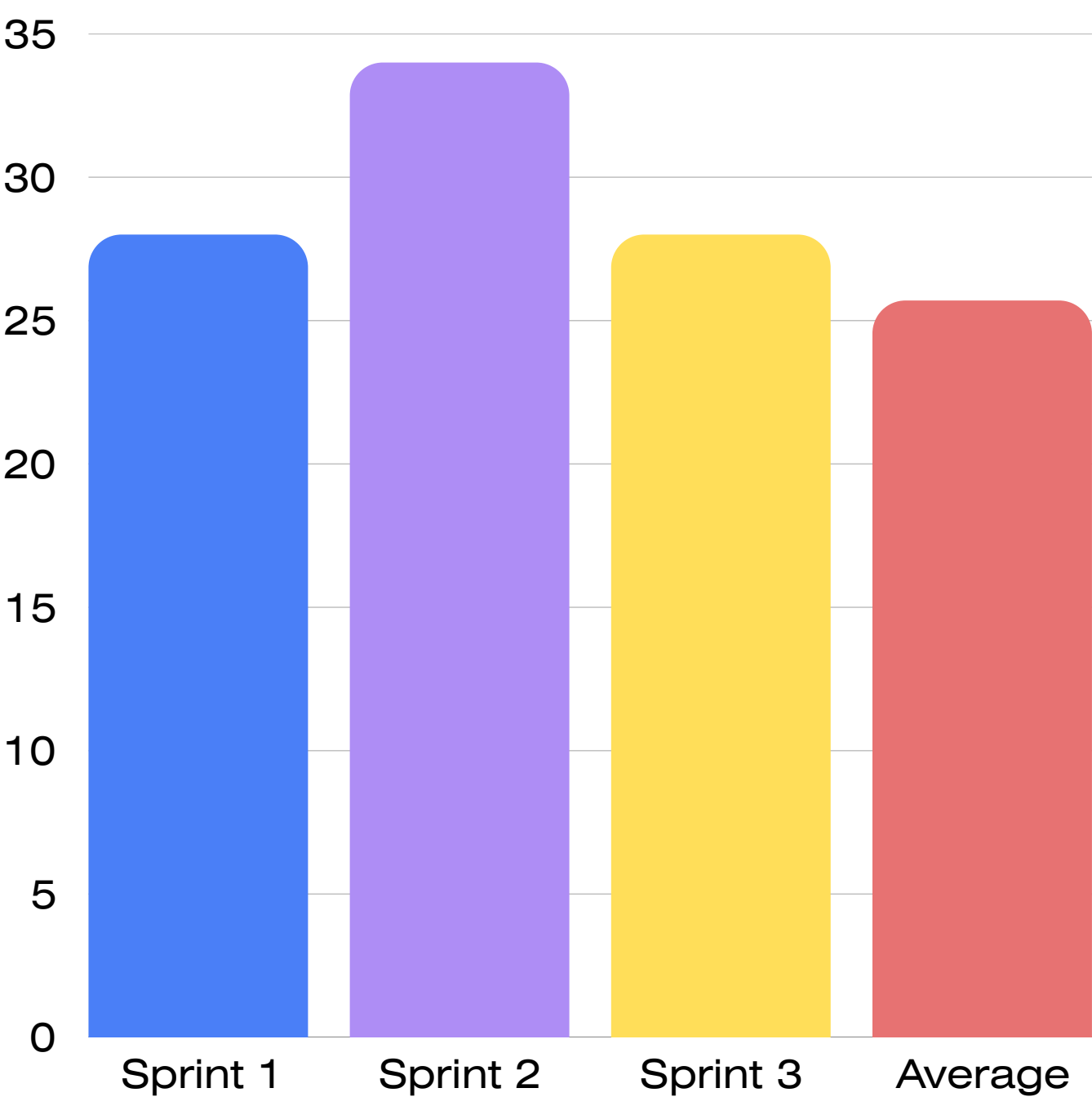
Team Velocity

Total velocity of sprint 1: 28

Total velocity of sprint 2: 34

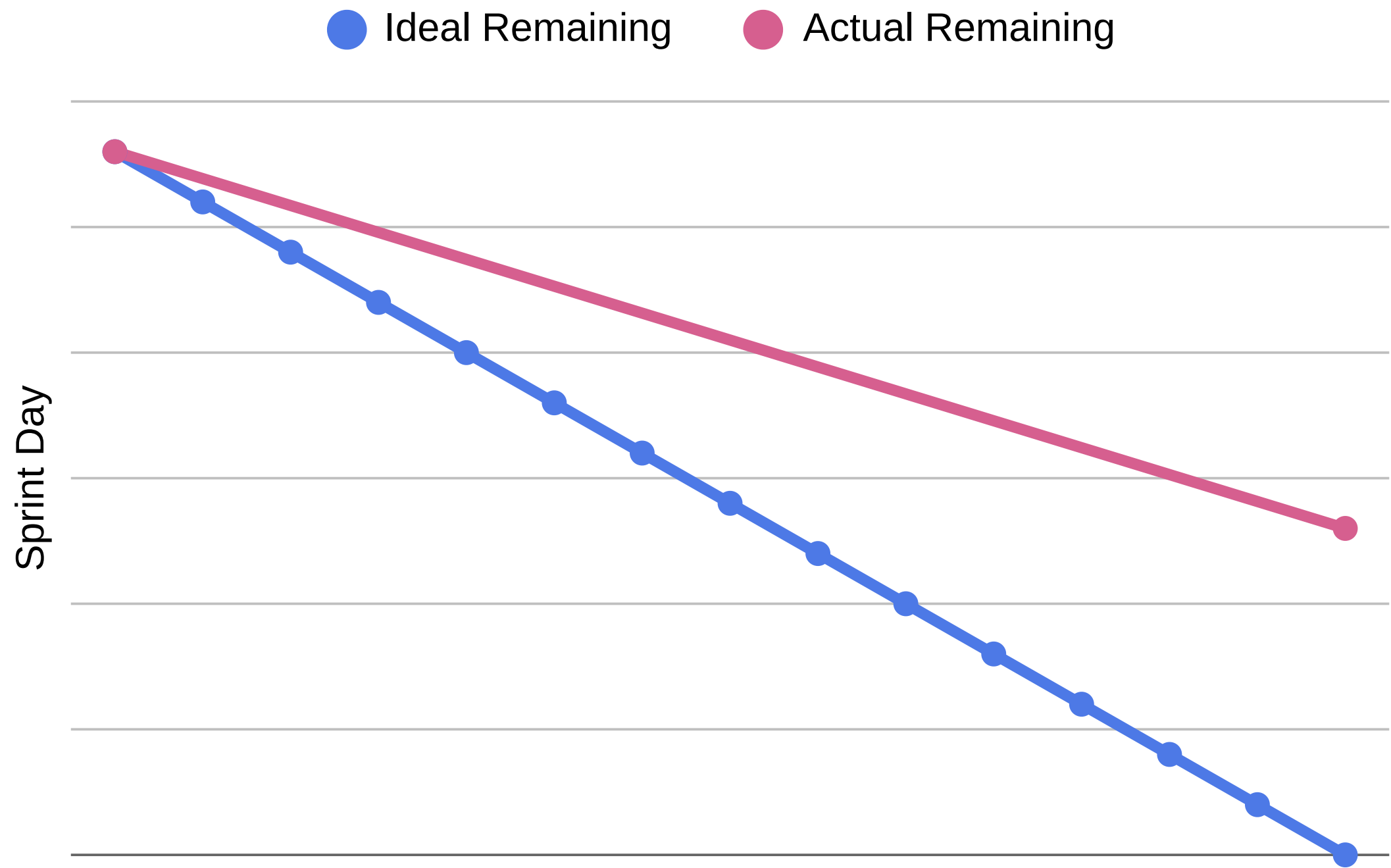
Total velocity of sprint 3: 15

Average: 25.7



BURNDOWN STATUS

Confirmed Metrics	
Committed Points	28
Delivered Points	15
Carried Over	13
Velocity %	53.6



COMPLETED VS COMMITTED

Sprint 1

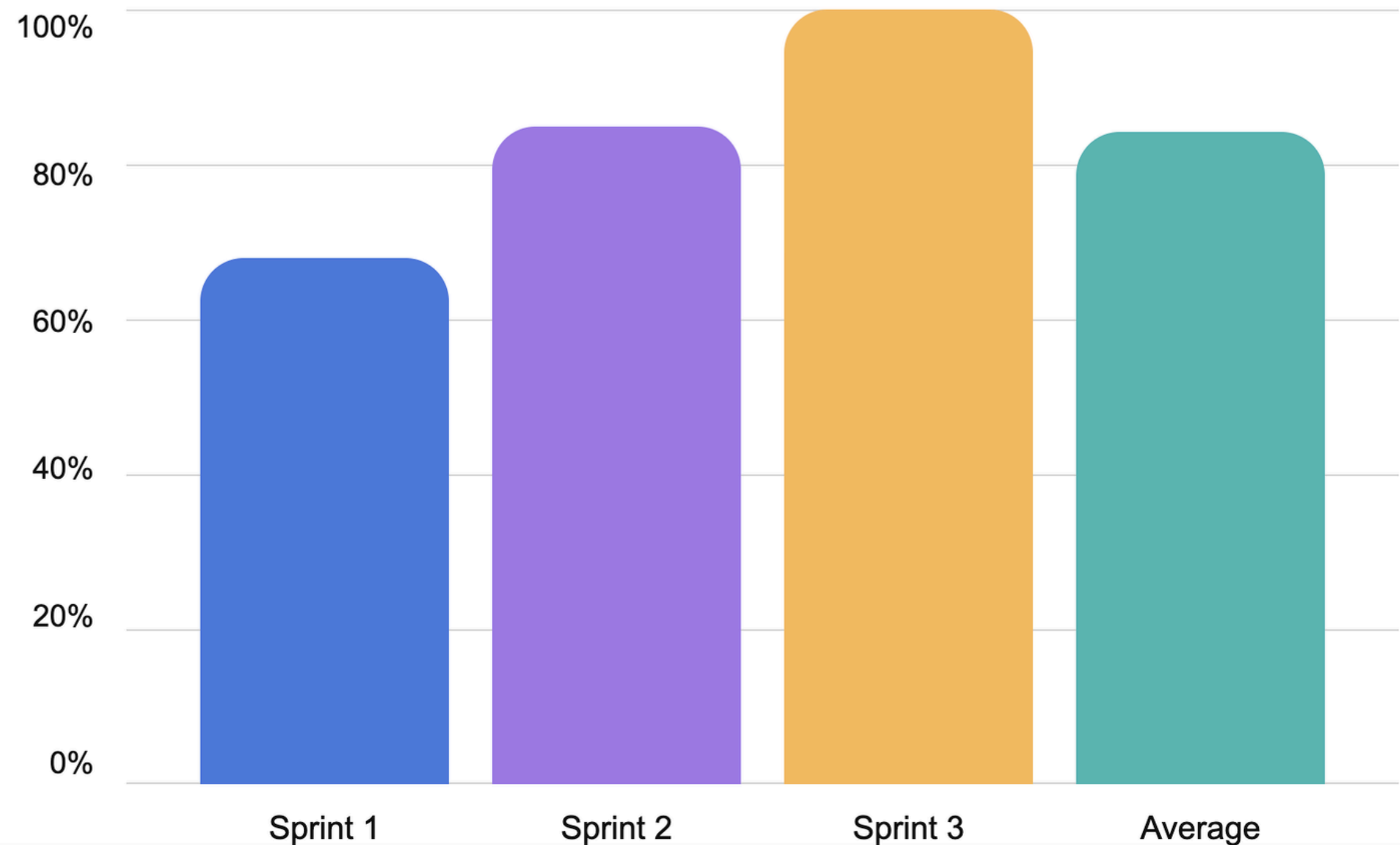
Committed: 41
Completed: 28
Ratio: 68%

Sprint 2

Committed: 40
Completed: 34
Ratio: 85%

Sprint 3

Committed: 28
Completed: 28
Ratio: 100%



RETROSPECTIVE

WHAT WENT WELL

Earlier start on documentation and slides

Better collaboration between co-owners on shared tasks

Team members supported each other when blocked

CHALLENGES

Ticket statuses weren't always kept up to date on the board

Workload wasn't always evenly distributed across the team

WIKI PAGE LINK

DEMO



THANK YOU

15 May, 2026